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1. Define cloud computing and explain four characteristics.

1. Defination: cloud computing means using computing services like storage, servves and software through the internet instead of maintaining everything on our own computer

Example: Google drives stores oure files online so we can access them from anywhere.

four characteristics:

1. On-demand service: we can use resources whenever we need them.

2. Broad network access: we can access cloud service using moblies, laptops, and computers.

3. Resource pooling: one provider shares computing resource among many users

4. Rapid elasticity: Resources can be increased or decreased according to demand.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution and modern cloud platforms.

Ans: Cdoud computing developed step by step:

1. mainframe computers: large computers were shared by many users

2. personal computer: people started using individual computers

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3. Client-server computing :- computers recevied Services from central computers

4. Virtuligation :- one physical server could run multiple virtual machines.

5. Internet growth : people user could access applications and information online.

6. Cloud computing :- servers, storage, and software machines platfrom available over the internet

Conclucion :- These development led to modern cloud platfroms such as Aws, mircrosoft Azure, and Google cloud.

3. Explain server Virtuligation and differentiate it from cloud computing.

Ans :-

Server virtualigation :- It is a technology that divides one physical Server into multiple virtual servers

Example :- one physical computer runs three virtual machines for different taskes

Advantages

- Reduces hardware cost.
- uses resource efficiently
- Runs multiple operating systems
- makes server marragment easier.

Difference:

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virtulization	cloud computing
i) It is a technology.	i) It is a service delivary model
ii) Creates virtual machine.	ii) provides computing service over the internet
iii) Can work without cloud Services	iii) may use virtuligation to provide service
iv) Ex: vmware	Ex: Aws.

4. Explain the role of data centers in cloud computing

Ans: A data center is a place where many server storage devices, and networking equipment are kept.

main role:

1. Data storage: Stores, files, photos, and databases
2. Application hosting: Runs websites and application
3. Networking: connects users to cloud services
4. Security: uses Security systems to protect data
5. Backup: keeps copies of important data for recovery
6. Availability: Helps services run reliably with backup power and equipment.

Ex: when you upload a photo to Google drive, it stored on Google cloud infrasturucture in data centers.

5. Differentiate Iaas, Paas, Saas, and Xaas with example:-

Ans:-

1. Iaas - Infrastructure as a service
   provide virtual servers, storage, and networking
   Example: Amazon EC2
   Easy meaning: Rent computer infrastructure.

2. paas: platform as a services
   provides a platform to develop, test, and run applications
   Ex:- Google App engine
   Easy meaning: Get a platform to build software

3. Saas - Software as a service
   provides ready to-use software through the internet.
   Example: Gmail
   Easy meaning: Use software without installing or managing its servers

4. Xaas - Anything as a Serice
   provides different types of IT Services over the internet
   Ex:- Dropbox offers cloud storage
   Easy meaning: Get IT serices whenever required.

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